Circuit investigation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use measurements to test a circuit model safely.

SUCCESS CRITERIA - I can...

1. Draw and predict before powering a circuit.
2. Place ammeters in series and voltmeters in parallel.
3. Compare measured and predicted values using evidence.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “ammeter” and “voltmeter” to explain your model or result. Sentence starters are optional.

ammeter: an instrument placed in series to measure current

voltmeter: an instrument placed across two points to measure voltage

trial: one run or repetition of a test

01 Warm-up: bring back an earlier idea

A circuit has 3 identical series resistors. Add two more at unchanged ideal supply voltage.
Compare new current with the original.

Word help: current = rate of charge flow

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

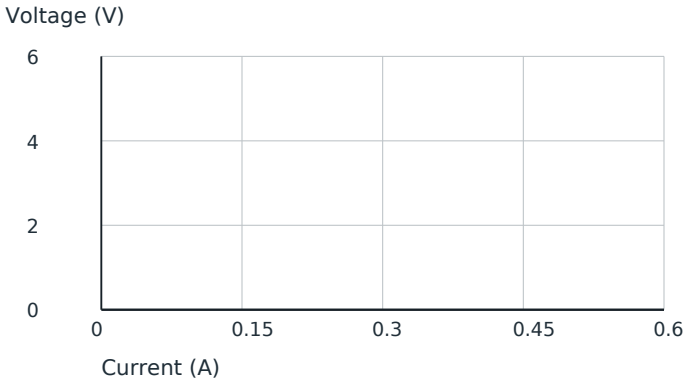
02 Worked example: explain the model

Reference: V = IR.
Use approved low voltage. Ammeter in series; voltmeter across a component. Disconnect before rewiring; never put an ammeter across the supply.

Worked reasoning: Supplied measurements show V=2, 4, 6 V and I=0.2, 0.4, 0.6 A. Each V/I is 10 ohms. This supports an ohmic model over the measured range.

Your turn: Plot V against I and mark the slope.
State why the data do not prove the model at every possible temperature or voltage.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



Marzano check - circle: ① ② ③

Circuit investigation

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

On a resistor circuit sketch, add an ammeter to measure loop current and a voltmeter to measure the resistor’s drop.

Sentence starter: The … in my model represents … . Its label shows … .



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Apply the physics

A lab trial records 5.8 V across a nominal 12 Ω resistor and 0.46 A through it.
Calculate measured resistance V/I and percent deviation from 12 Ω.
Use the table to separate prediction, measurement and a plausible uncertainty source.

Sentence starter: I predict … . My model supports this because … .

Nominal R (Ω)	Measured V (V)	Measured I (A)	V/I (Ω)
12	5.8	0.46	____

Marzano check - circle: ① ② ③

05 Explain your reasoning

A reading disagrees with your prediction. List two checks before replacing the model or deleting data.

Sentence starter: I predict … . My model supports this because … .

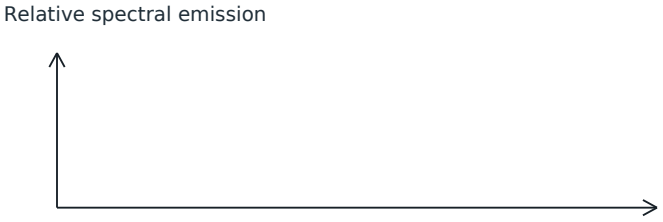
Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

An ideal blackbody has peak wavelength 580 nm.
Use λ_{max} $T = 2.90 \times 10^{6}$ nm·K.
Estimate temperature.
Show how the thermal peak changes.

Word help: blackbody = an ideal absorber/emitter used in a thermal-radiation model

Sentence starter: I found … [unit]. This value means … in this situation.



Wavelength (label units and peaks)

Marzano check - circle: ① ② ③

Circuit investigation

antocciphysics

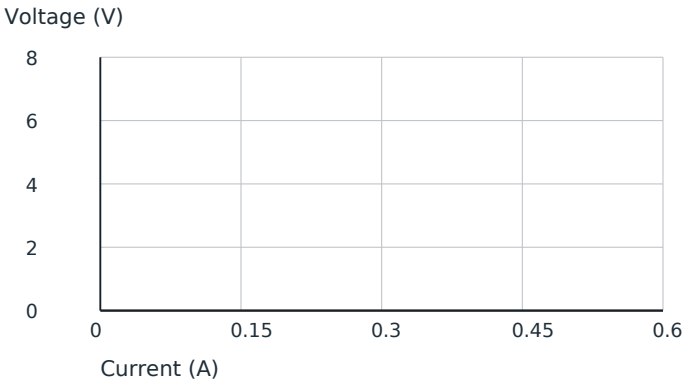
Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

Design a three-point V-I test.
State what to control and what pattern would support constant resistance.

Sentence starter: My design shows … . It meets the requirement by … .



Marzano check - circle: ① ② ③

08 STRETCH: test the model

Two readings are 3.0±0.1 V and 0.30±0.02 A.
Find central resistance and worst-case bounds.

Sentence starter: I found … [unit]. This value means … in this situation.

Quantity	Lower bound	Central value	Upper bound
Voltage (V)	____	____	____
Current (A)	____	____	____
Resistance (Ω)	____	____	____

Marzano check - circle: ① ② ③